

Lifestyle Factors Impact on Sleep Quality

By Team RSA (Data-Centric Project): Ricky Bertone, Alisha Srivastava, Sara Evans

Medium Article: [Link](#)

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Sara Evans · 12 min read · Just now



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Problem:

For this project, our team wanted to explore how different lifestyle factors impact our sleep, specifically focusing on stress, screen time, and academic workload. Our project investigates how these various factors affect sleep quality, sleep duration, and disruptions during the night. As college students, we frequently experience varying levels of stress, fluctuating screen time use, and heavy academic workloads, all of which may contribute to changes in our sleep quality.

By tracking these factors over the course of five weeks, we aimed to determine whether certain habits have a significant impact on the quality of our sleep and help us better understand how our own routines affect our ability to rest properly.

We aimed to answer the following questions:

- Is there a relationship between daily stress levels and how often we wake up during the night?
- Does eliminating screen time within one hour before bed improve sleep quality?
- Does the number of hours spent studying or doing schoolwork each day affect the total sleep duration we achieve?

To attempt to answer these questions, we collected both manual and automatic data sources. Each team member recorded daily lifestyle factors in their individual spreadsheets, including stress level (1–5), sleep quality (1–5), the number of times they woke up during the night, whether they used screens within an hour before going to bed (yes or no). The scale of 1–5 was used for stress level and sleep quality, with 1 indicating the lowest level of stress and poorest sleep quality, and 5 represented the highest level of stress and best level of sleep quality. Additionally, we tested the effect of screen time by avoiding screens an hour prior to bed for one week and then using them within an hour before bed the next week, alternating between the two throughout the 5 weeks. In addition to our manual data sources, we used the Apple Health app to automatically record our nightly sleep duration. Our data collection began on 3/31/2025 and ended on 5/2/2025, excluding weekends.

Solution:

Collecting the data above supports answering our questions as it allows us to track individual lifestyle factors, identify key sleep patterns, and assess sleep quality. By recording stress levels, screen time, and academic workloads, we were able to explore the relationship between these lifestyle habits and different aspects of our sleep including sleep quality, sleep duration, and number of times we woke up during the night. We visualized our data primarily through heatmaps and individual scatterplots to help us identify potential correlations and trends.

However, our data collection approach contained several limitations in regards to providing accurate answers to our questions. A significant limitation includes the potential bias from self-reporting the majority of our data. For example, forgetting to track our data immediately after waking up or completing our academic work for the day, may have resulted in inaccuracies within our data. Additionally, we collected stress level and sleep quality using a subjective scale of 1–5, which may have led to inconsistencies, as each team member could be interpreting the scales differently based on their own perceptions. With subjective data, it's difficult to come to conclusive answers and create valid comparisons when our ratings are based on different standards for what is considered high stress or poor sleep quality.

Data Pipeline:

For our data collection, we collected daily data on stress level (1–5), sleep quality (1–5), the number of times they woke up during the night, whether they used screens within an hour before going to bed (yes or no) which were self recorded by each team member in their individual spreadsheet.

Additionally, we collected sleep duration using the Apple Health app. For our data extraction process, we combined our individual spreadsheets containing our self-reported data into a single spreadsheet. In addition, we exported our sleep duration data from the Apple Health app and converted it into a CSV file using a Python script. These CSVs were then manually transferred into our main spreadsheet that contained the rest of our data. For our data analysis, we cleaned and analyzed our data using a combination of R and Python. We mainly used these tools to standardize our variable names and generate visualizations to present our findings.

Tools Used:

- Tracking Stage: Microsoft Excel, Apple Health
- Data Visualization: Python, R Studio
- Infographic: Figma
- Presentation: Google Slides
- Final Blog Post: Medium

The only challenge we faced was that by using individual spreadsheets to track our data, we named our variables differently which made it a bit difficult when trying to analyze our data. We fixed this by manually changing the names to make them the same.

One of the questions that we updated during the project was the second question we answered regarding academic workload. Originally, we were going to track how much time was spent in class during the day, but we realized that it may be uncorrelated due to time doing work outside of class as well. We updated the question to include both time spent in class, working on assignments, as well as time spent studying to get a total time (in hours) spent working on academics during the day. This was a successful change as it allowed us to get a more accurate representation of the feature we were attempting to measure without making a completely new question.

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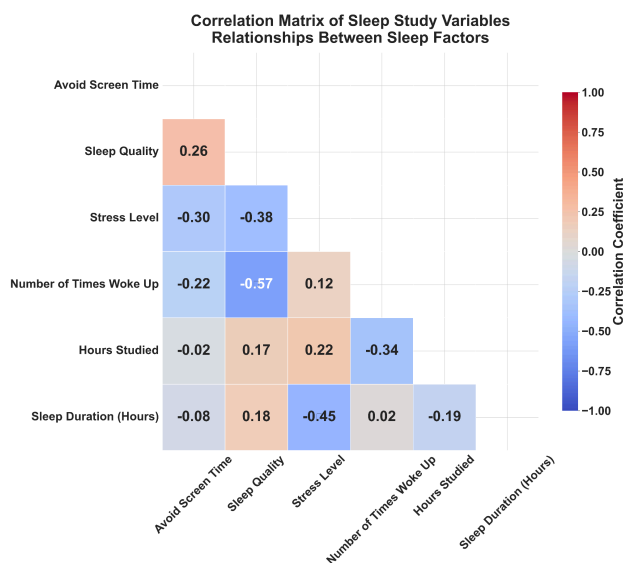
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Insights:

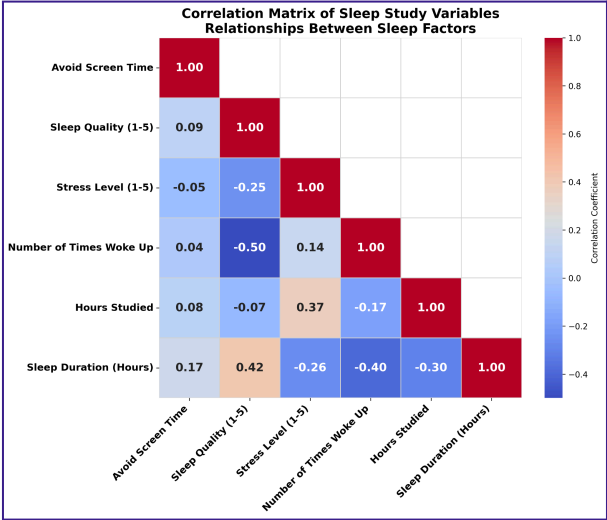
Milestone Two (Intermediate Analysis):



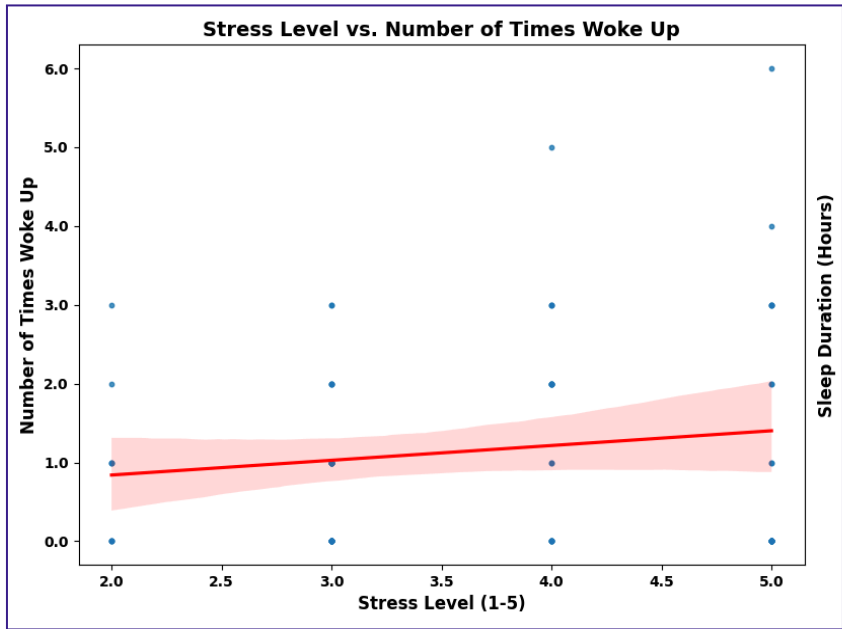
Our major Milestone 2 Findings were that waking up at night is the strongest negative factor for sleep quality with correlation at -0.57. Avoiding screen time (ST) helps, but only a little (+0.26), and avoiding ST reduces stress and nighttime waking as well, with correlations at -0.30 and -0.22 respectively. Some of the questions we wanted to answer could be supported by this data. For example, higher stress levels correlate with worse sleep. Collecting more data would allow us to confirm any question that might not be completely answered. For example, we do not have sufficient data to prove that preventing screen time usage an hour before bed has a significant effect on

our sleep quality. Therefore, we expected our correlations to get stronger with more data.

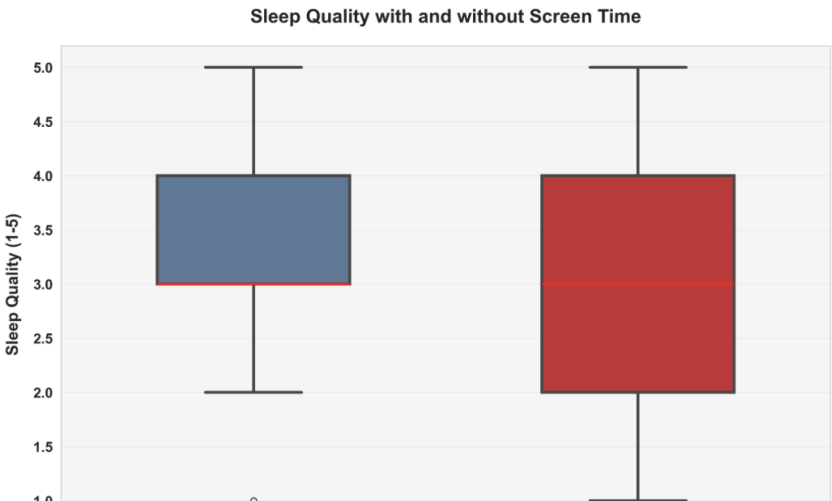
Milestone 3:



On Milestone 3, we thought our correlations would get stronger with an additional 3 weeks of data. However, in Milestone 3, we had weaker correlations overall. From the graph, you can see compared to Milestone 1, most of the correlations have moved closer to zero. We think the reasons for this are that the original dataset at Milestone 2 may have had outliers or may have been in a stage where our behaviors supported the hypotheses more. We also know that we had a small sample size, so our correlations may have been amplified.

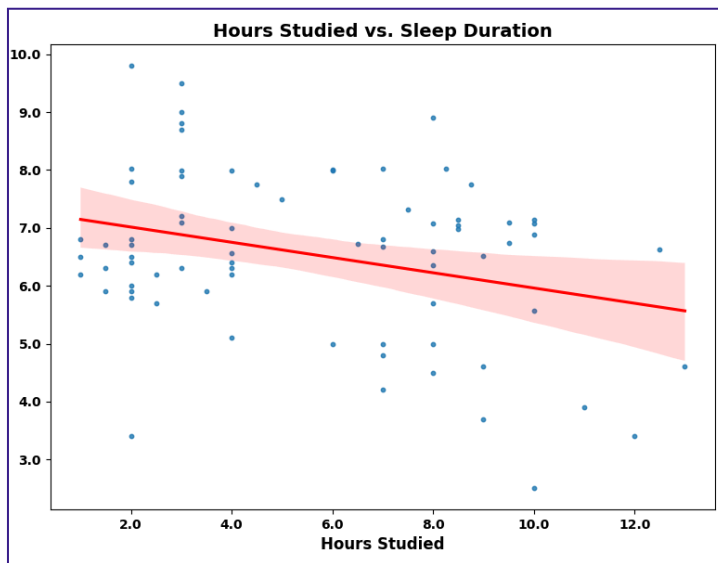


Our first question asked, how are stress level and night wakings related? Higher levels of stress increases with the number of times we wake up at night. This trend suggests that people with higher stress tend to wake up slightly more often. You can observe from the graph that there is a weak correlation with a wide confidence interval. Hypothesis is not supported here.



Our second hypothesis asked about screen time and sleep quality. Milestone 2 Correlation was 0.26, which moderately positively. On Milestone 3, the correlation became 0.09, which is a very weak positive. Adding

more data increased noise over time.



With Study Time vs. Sleep Duration, we found that there was a negative trend. Therefore, we found that increased study hours are correlated to reduced sleep duration. This tells us that academic workload might be cutting into rest time. This means that our hypothesis that more study hours will reduce the amount of sleep was supported by our study.

Our final findings are that waking up at night is strongly negatively correlated to sleep quality. Higher stress correlates with shorter and lower-quality sleep, as seen in Milestone 2 and 3. After collecting more data, we found that sleep duration and sleep quality now have a stronger positive link.

Reflections:

Each team member was asked to reflect on their tracking experience and answer the following questions:

- What were your expectations from the tracking experience?
- Was there any highlight of the tracking experience?
- Was there a positive outcome from the tracking experience?
- Was there a negative outcome from the tracking experience?
- If you were to do this project again, what would you change?

Alisha:

- In Milestone 2, I said that “having one additional thing to do everyday was a bit annoying because it was one more thing to remember, but the short duration of the task made it doable”, and by Milestone 3, I was just used to tracking my data. I was expecting to see

awful sleep and stress levels, and I did expect to see a limited number of times of waking up in the night since I usually sleep quite deeply.

- The highlight was that I tried to sleep better and as much as possible so that we had more results. I also think making the graphs was fun. I did end up sleeping more just due to being more conscious.
- I think it helped me be more conscious of the amount of time I was sleeping, and how much time I was spending on schoolwork. I think that was very important to pay attention to, because it impacts my health levels.
- I realized that I spend a significant amount of time studying and that correlates with the amount of stress I experience per day, and that school is a significant stressor in my life. I also realized that my poor sleep schedule is a consequence of the amount of hours I spent working on school work and my research. However, I am a college student so I understand that this is expected, but I'm looking forward to a break! Additionally,
- I would first include more automated variables. For example, I would like to include screen time and other numeric variables for better analysis. Second, I would like to address more of the confounding variables such as exercise which can contribute to deeper sleep or less screen time through the day, which could offset the effect of screen time.

Sara:

- My expectations from the tracking experience included being able to better understand how my daily habits impacted my sleep quality. By tracking factors related to my sleep, I hoped to identify any patterns that either contributed to getting a good night's sleep or led to a bad one.
- A highlight of the tracking experience was that I became more mindful of my daily habits and gave more consideration as to how they may impact my sleep.
- A positive outcome from the tracking experience was that I became more conscious of my sleep habits overall. Before tracking, I didn't think about how my behaviors throughout the day might affect my sleep, but since tracking I've realized that certain habits such as screen time before bed may negatively impact my quality of sleep..
- The main negative outcome from my tracking experience would be that I often became too aware of any behavior that might affect my sleep. For instance, on nights when I avoided screens an hour before going to sleep, I became focused on not falling asleep too soon, which made it more difficult to fall asleep normally.
- If I were to do this project again, I would increase the length of our study as I believe it would help provide us with clearer results. Additionally, I believe that increasing our sample size beyond three participants would also help provide us with more accurate results.

Ricky

- I was expecting to learn more about my habits and what things I do that directly affect my sleep. I had expected that I slept less than 8 hours on average each night and that I woke up at least once per night usually.
- The highlight of my tracking experience was probably the improvement I saw in my daily life. Waking up and inputting the data from the night before had helped me to become more mindful of my overall sleep health as well as my daily stress levels so I could have the best sleep that I could.
- The positive outcome of the tracking experience was that I began to understand what factors affected my sleep pretty quickly. Another positive outcome of the experience was that I learned that I would feel much more refreshed in the morning and easily ready to get out of bed on days where I did not use screen time before bed, even if I slept less than on a night where I did use a device.
- One negative outcome was that I learned just how bad my current sleep experience is. I did not expect the number of wakeups during the night to be so high until I began tracking it seriously. Although I had expected not to have an average of 8 hours of sleep a night, I did not expect that the numbers would often be closer to 5 hours of sleep.
- If I were to do this project again, I would do a longer study. I would also increase the duration of the intervals for the study, like doing a month or multiple months straight of removing screen time before bed in order to see if there are long term effects. I would also use more automatic data tracking through a smart watch or other monitoring device to achieve more accurate data, as there were likely biases with self reporting in our data.

Contributions:

Alisha:

- Milestone 1
 - First, I helped ideate what the project should be and what the research questions should be. Based on that, we thought about what could realistically be measured and quantified, and then I helped design the plan for data collection based on the questions we had: deciding what can be automated and what can't be, and what the tradeoffs are for each situation.
- Milestone 2
 - First, everyone sent me their data. I standardized the data by 1) fixing the date formatting, 2) fixing the ordering of the columns to stack the data in long format, 3) creating a column to identify which study participant (richard, sara, or i) each data point came from, and 4) standardizing the column names. I also created encoded variables where needed upon the categorical data, and where we had numeric data, I rounded it to one decimal point to better plot it. I also created our visualizations, ran the t tests, interpreted the data, and created the infographics, and created our correlation graphs which ended up being our major final analysis. I also then wrote the interpretations and final analysis of our study.
- Milestone 3

- First, everyone sent me their data. I standardized the data again by 1) fixing the date formatting, 2) fixing the ordering of the columns to stack the data in long format, 3) creating a column to identify which study participant (richard, sara, or i) each data point came from, and 4) standardizing the column names so that I could add the data from the additional weeks. I also created encoded variables where needed upon the categorical data, and where we had numeric data, I rounded it to one decimal point to better plot it. I also created our visualizations, ran the t tests, interpreted the data, and created the infographics, and created our correlation graphs which ended up being our major final analysis. Then, I compared our Milestone 2 data to our Milestone 3 data and wrote about that. I also wrote the final findings/conclusions, and wrote about the limitations of our study, small sample size for example. I created the analysis section of our slideshow as well.

Sara:

- Milestone 1
 - I focused on developing our research questions and determining the overall analysis plan, including what variables we needed to track based on our project's goals. I also helped to write the project summary, highlighting the purpose and significance of our project.
- Milestone 2
 - I wrote our project statement and refined our research questions based on what we learned from our first few weeks of data collection. I contributed to building our data pipeline and assisted in interpreting our visualizations.
- Milestone 3
 - I wrote the problem and solution statements, helped refine the data pipeline for our completely collected data, and contributed to creating and interpreting our visualizations.

Ricky:

- Milestone 1
 - I was responsible for data collection, developing research questions, as well as creating the timeline to be used for Milestone 2. I also did the team's contribution statement.
- Milestone 2
 - In milestone 2 I continued my data collection, and performed a small amount of analysis of our visualizations. I also wrote the contribution statement and created the timeline for milestone 3.
- Milestone 3
 - I wrote the contribution statement and created the timeline for the final presentation as well as the final report blog post.